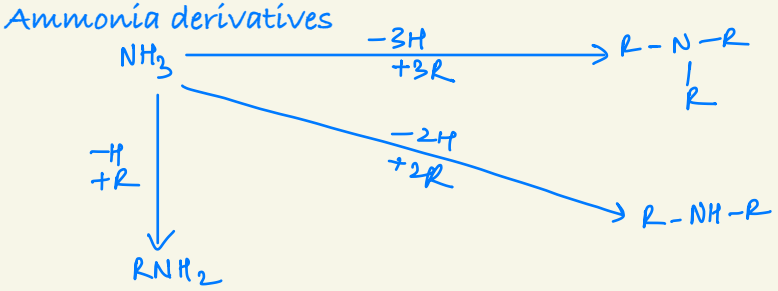
12th hacker series

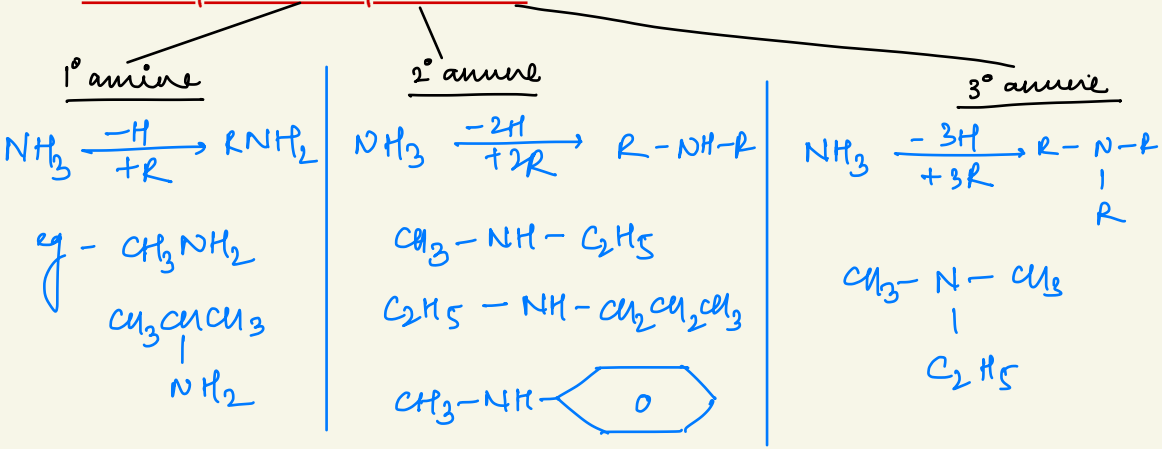
Amines

Amines

Ammonia derivatives



Classification of Amines



Nomenclature

Primary Amine − RNH_2

c/N → alkylamine

IUPAC → alkanamine

eg − CH_3NH_2

c/N − methylamine

IUPAC − methanamine

Word root + 1° suffix + 2° suffix

alkane + amine ⟶ **alkanamine**

eg - $CH_3\overset{2}{C}H(\overset{1}{C}H_3)\overset{3}{C}H_3$ with NH_2 on C-2

IUPAC - propan-2-amine

Naming and structures of amines

	C/N	IUPAC
① $CH_3CH_2NH_2$	ethylamine	ethanamine
② $CH_3CH_2CH_2NH_2$	N-propylamine / n-propylamine	propan-1-amine (or) propanamine
$H_3\overset{1}{C}-\overset{2}{C}H-NH_2$ with $\overset{3}{C}H_3$	Isopropylamine	propan-2-amine

nomenclature for secondary amine

$R-NH-R$

$R-NH-R$

C/N - dialkylamine

IUPAC - N-alkylalkanamine

eg $CH_3-NH-CH_3$

C/N - dimethylamine

IUPAC - N-methylmethanamine

$R-NH-R'$

C/N - alkylalkylamine

IUPAC - N-alkylalkanamine

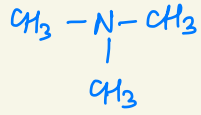
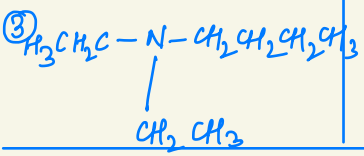
CH ↓ CTN

eg $\overset{\text{methyl}}{CH_3}-NH-\overset{\text{ethyl}}{C_2H_5}$

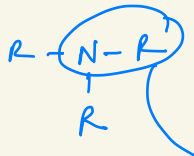
C/N - ethylmethylamine

IUPAC - N-methylethanamine

Naming and structures of amines

	CN	IUPAC
① $CH_3-NH-C_2H_5$	ethyl methylamine	N- methylethanamine
② 	Trimethylamine	N,N- dimethyl methanamine
③ 	N,N- diethylbutylamine	N,N- Diethylbutan-1-amine

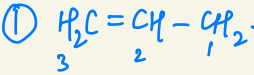
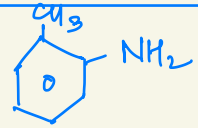
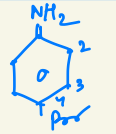
Naming of tertiary amines

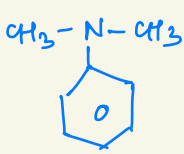


CN - trialkylamine

IUPAC - N,N - dialkylalkanamine

Naming and structures of amines

① 	Allylamine	prop-2-en-1-amine
② $NH_2-(CH_2)_6-NH_2$	hexamethylene diamine	hexane-1,6-diamine
③ 	Aniline	Aniline or Benzenamine
④ 	ortho toluidine	2-methylaniline
⑤ 	p-bromoaniline	4-bromoaniline

(6) 

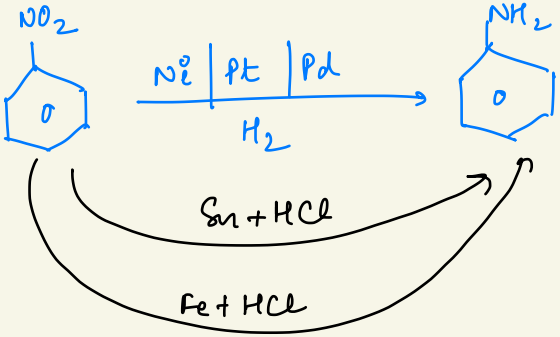
N, N - dimethylaniline

N, N - dimethyl benzenamine

Methods of preparation of amines

Reduction of nitro compounds

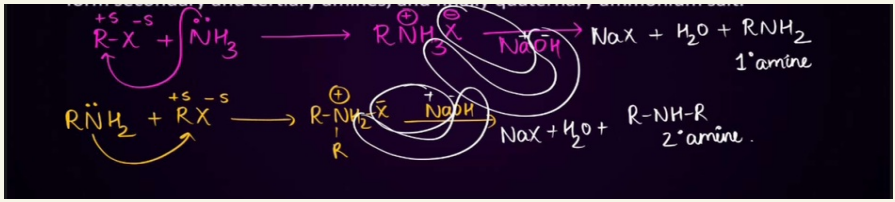
nitro compound are reduced to amine by passing hydrogen gas in the presence of finely divided nickel, palladium or platinum, and also by reduction with metals in acidic medium. Nitroalkanes can also be similarly reduced to the corresponding Alkanamines

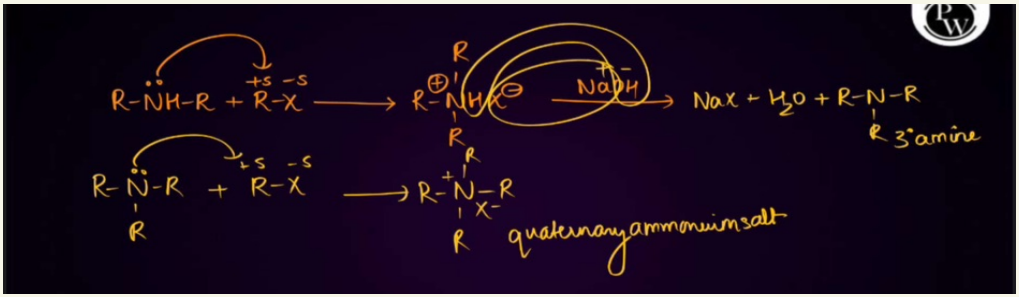


- Reduction with iron scrap and hydrochloric acid is preferred because FeCl2 formed gets hydrolysed to release hydrochloric acid during the reaction.Thus, only a small amount of hydrochloric acid is required to initiate the reaction.

Ammonolysis of alkyl halide

The reaction is carried out in a sealed tube at 373 K. The primary amine thus obtained behaves as a nucleophile and can further react with alkyl halide to form secondary and tertiary amine and finally quaternary ammonium salt.





Ammonolysis has the disadvantage of yielding a mixture of primary, secondary and tertiary amines, and also a quaternary ammonium salt. However, primary amine is obtained as a major product by taking large excess of ammonia. The order of reactivity of halides with amine is

$RI > RBr > RCl$

Reduction of nitriles

On nitriles reduction with lithium aluminium hydride or catalytic hydrogenation produce primary amine . This reaction is used for ascent of amine series. I.e, for preparation of amine containing one carbon atom more than the starting amine

$$RCN \xrightarrow[Ni/Al/Pd,\ H_2]{LiAlH_4} RCH_2NH_2$$

$Na/Hg + C_2H_5OH$

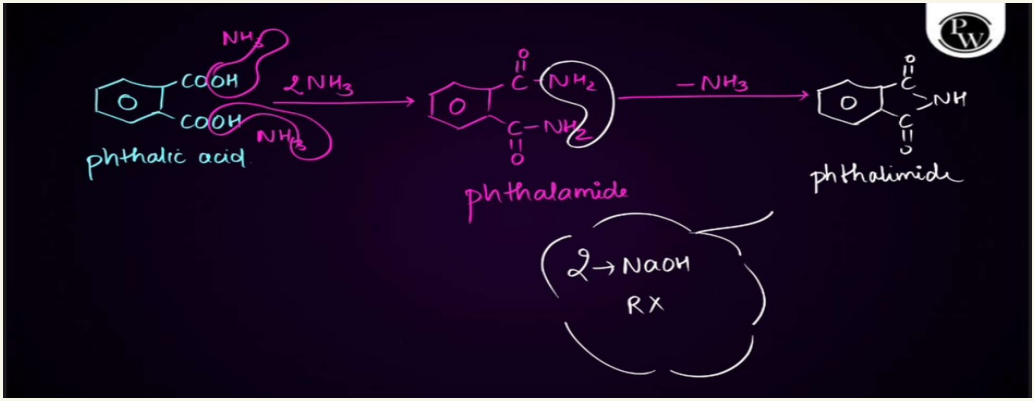
Reduction of amides

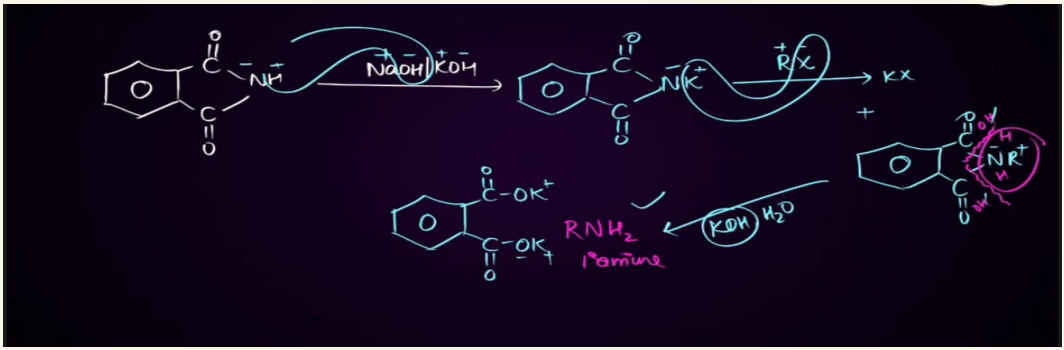
Reduction of amide with lithium aluminium hydride yield amines

$$R-C-NH_2 + 4[H] \xrightarrow{LiAlH_4} R-CH_2-NH_2$$

Gabriel phthalimide synthesis

Gabriel synthesis is used for the preparation of primary amines. Phthalimide on treatment with ethanolic potassium hydroxide forms, potassium salt of phthalimide which on heating with alkyl halide, followed by alkaline hydrolysis produces the corresponding primary amine . Aromatic primary amine cannot be prepared by this method because aryl h undergo nucleophilic substitution with the anion formed by Phthalimide





Hoffmann bromamide degradation reaction

- Hoffmann developed a method for preparation of primary amines by treating an amide with bromine in an aqueous or ethanolic solution of sodium hydroxide. In this degradation reaction, migration of an alkyl or aryl group takes place from carbonyl carbon of the amide to the nitrogen atom. The Amine so formed contains one carbon less than that present in the amide.

$$RCONH_2 + Br_2 + 4NaOH \rightarrow RNH_2 + Na_2CO_3 + 2NaBr + 2H_2O$$

Physical properties of amines

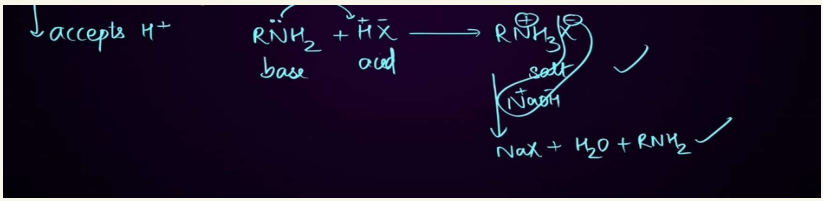
- the lower aliphatic amine are gases with fishy odour. Primary amines with three or more carbon atoms are liquid and still higher ones are solid. Aniline and other Arylamines are usually colourless but get coloured in storage due to atmospheric oxidation.

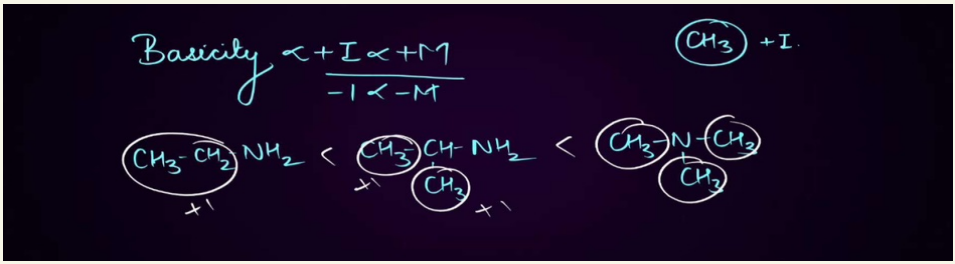
- lower aliphatic amines are soluble in water because they can form hydrogen bonds with water molecules. However, solubility decreases with increase in molar mass of Amines due to increase in size of the hydrophobic alkyl part. Higher amines are essentially insoluble in water.
- amines are soluble in organic solvent like alcohol, ether, and benzene. You may remember that alcohols are more polar than amines and form stronger intermolecular hydrogen bonds than amines

Chemical properties of amines

Basic Character of amines

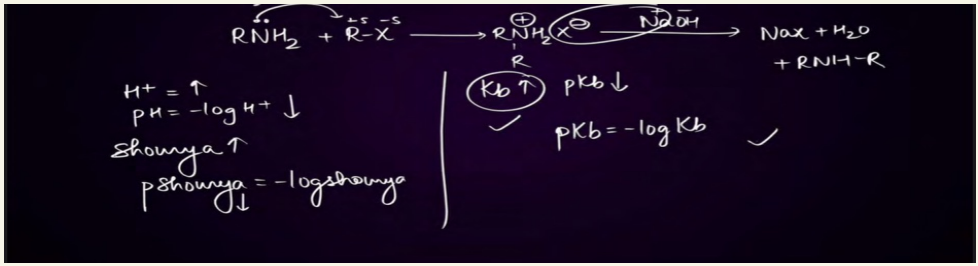
- amine is being basic in nature, react with acids to form salts





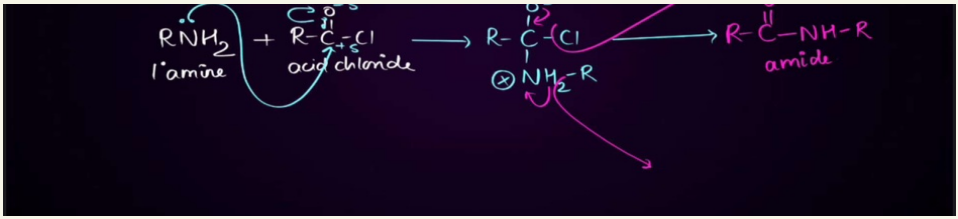
Alkylation

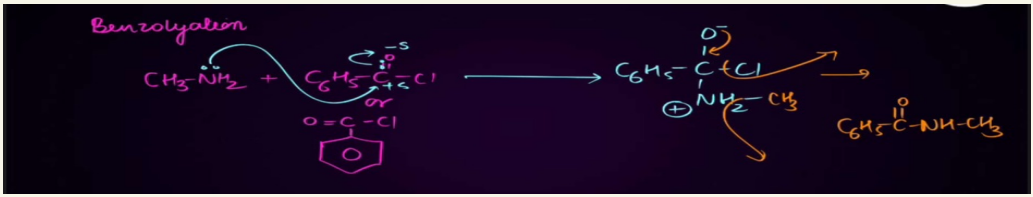
Amines undergo alkylation on reaction with alkyl halide



Acylation

- aliphatic and aromatic primary and secondary amines react with acid chloride anhydrides and esters by nucleophilic substitution reaction. This reaction is known as acylation





Carbylamine reaction

- aliphatic, and aromatic primary amines on heating with chloroform and ethanolic potassium hydroxide form isocyanides, or carbylamines which are foul smelling substances
- Secondary and tertiary amines, do not show this reaction
- This reaction is known as carbylamine reaction or isocyanide test and is used as a test for primary amines

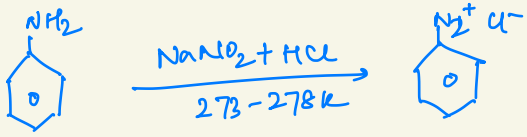
$$RNH_2 + CHCl_3 + 3KOH \rightarrow RNC + 3KCl + 3H_2O$$

Reaction with nitrous acid

- primary aliphatic amines react with nitrous acid to form aliphatic diazonium salts, which being unstable liberate, nitrogen gas quantitatively, and alcohols

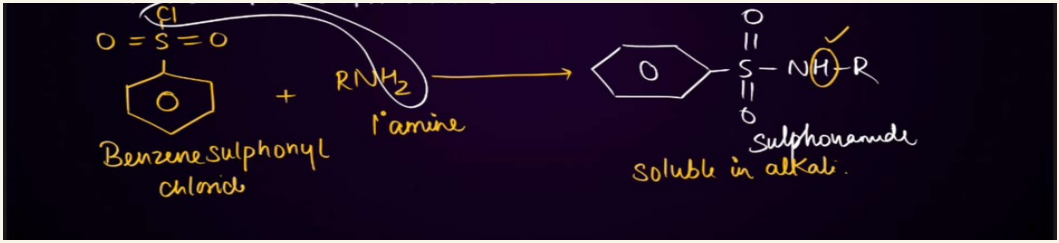
$$RNH_2 + HNO_2 \xrightarrow[HCl]{NaNO_2} R\,N_2^+\,Cl^- \xrightarrow{H_2O} R\text{-}OH + N_2 + HCl$$

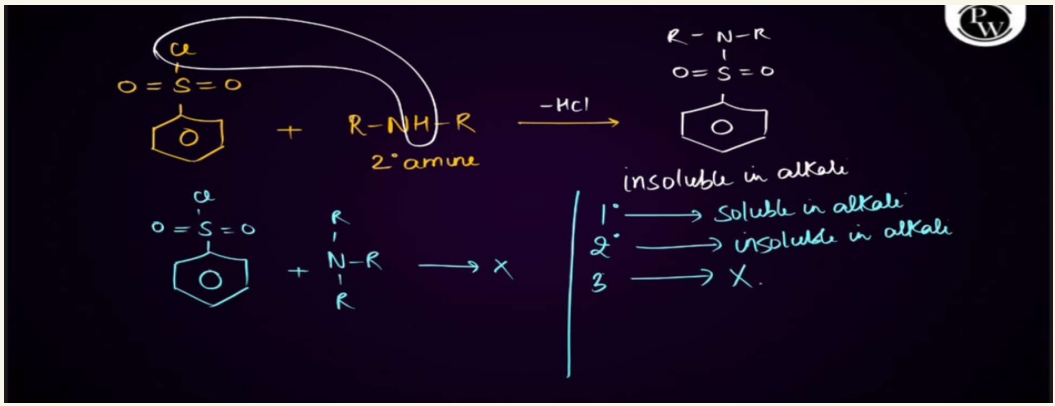
- aromatic amine react with nitrous acid at low temperature (273-278 K) to form diazonium salts, a very important class of compounds used for synthesis of a variety of aromatic compounds ☺

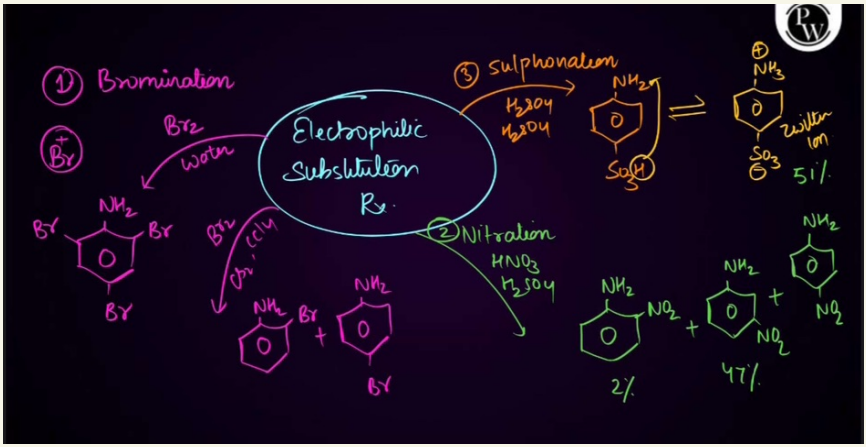


Reaction with arylsulphonyl chloride

- benzenesulphonyl chloride, which is also known as Hinsberg's reagent, reacts with primary and secondary amines to form sulphonamide



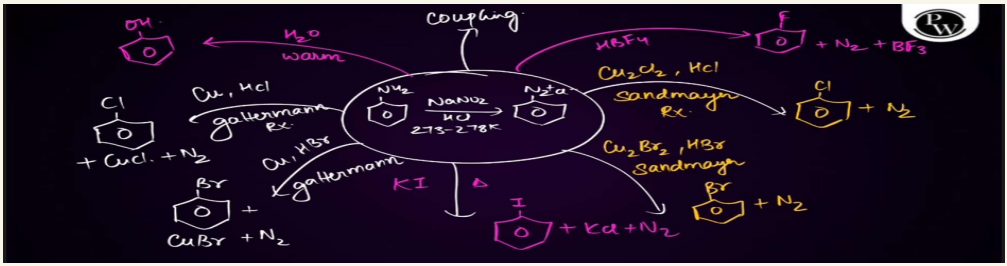


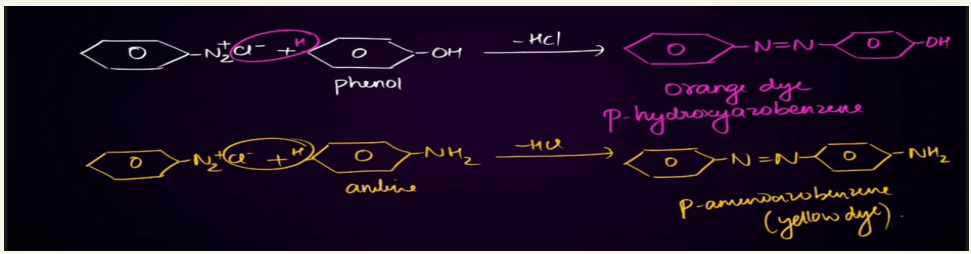


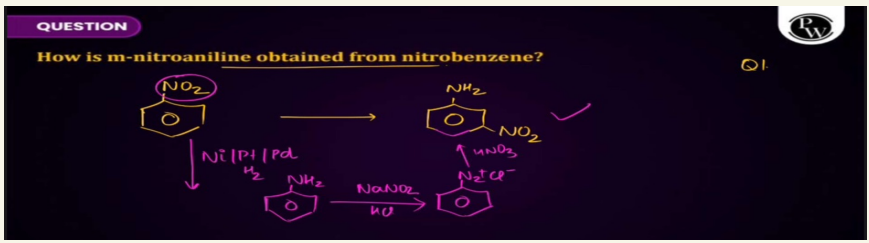
Diazonium salt

Introduction and method of preparation

- benzene diazonium chloride is prepared by the reaction of aniline with nitrous acid at 273- 278 Kelvin
- Nitrous acid is produced in the reaction mixture by the reaction of sodium nitrite with hydrochloric acid
- The conversion of primary aromatic amine into diazonium salt is known as diazotisation . Due to its instability, the diazonium salt is not generally stored and is used immediately after its preparation.
- Benzene diazonium chloride is a colourless crystalline solid
- it is readily soluble in water and stable in cold, but react with water when warmed
- it decomposes easily in the dry state
- Benzenediazonium flouroborate is water insoluble, and stable at room temperature







QUESTION

Out of CH_3NH_2 and $(CH_3)_3N$, which one has higher boiling point?

[Delhi 2014(C)]

Q2.

CH_3NH_2 > $(CH_3)_3N$

H ↑

H Bond ↑

QUESTION

Arrange the following compounds in increasing order of solubility in water:

$C_6H_5NH_2$, $(C_2H_5)_2NH$, $C_2H_5NH_2$

A B C

[Delhi 2014]

Q3

A < B < C

QUESTION

Why is an alkylamine more basic than ammonia?

[KVS]

Q4.

R = +I = EDG.

QUESTION

Arrange the following compounds in an increasing order of basic strengths in their aqueous solutions:

NH_3, CH_3NH_2, $(CH_3)_2NH$, $(CH_3)_3N$

[Delhi 2013, 12]

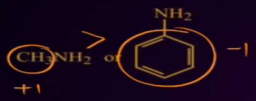
Q5.

2° > 1° > 3° > NH_3

gas 3° > 2° > 1° > NH_3

QUESTION

Which of the two is more basic and why?

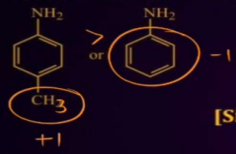


Q6.

[KVS; Foreign 2014]

QUESTION

Which of the two is more basic and why?



Q7.

[Similar to CBSe Sample Paper 2018, Foreign 2014]

QUESTION

Arrange the following increasing order of basic strength:

$C_6H_5NH_2$, $C_6H_5NH(CH_3)$, $C_6H_5N(CH_3)_2$

< (+I) < (+I, +I)

[Delhi 2014]

Q 8

QUESTION

Why do amines react as nucleophiles?

Lone pair

Nucleus loving

n + P ⊕

Nu⁻

QUESTION

Write chemical equations for the following conversions:

(i) $C_6H_5 - CH_2 - Cl$ into $C_6H_5 - CH_2 - CH_2 - NH_2$

(ii) $CH_3 - CH_2 - Cl$ into $CH_3 - CH_2 - CH_2 - NH_2$

[Delhi 2012(C)]

Q.

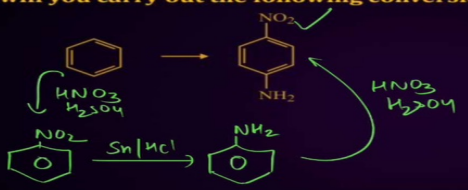
$C_6H_5-\overset{+}{C}H-\overset{-}{Cl} \xrightarrow{KCN} C_6H_5-CH-CN \xrightarrow[H_2]{Ni|Pt|Pd} C_6H_5-CH_2-CH_2-NH_2$ ✓

KCl +

(ii) $CH_3-CH_2-Cl \xrightarrow{+KCN} CH_3CH_2CN \xrightarrow[H_2]{Ni|Pt|Pd} CH_3CH_2CH_2NH_2$

QUESTION

How will you carry out the following conversion?



QUESTION

Identify A and B in each of the following processes:

C≡N

CH_3CH_2CN

(i) $CH_3CH_2\overset{+}{Cl} \xrightarrow{\overset{+}{N}\overset{-}{C}N} A \xrightarrow[Ni/H_2]{\text{reduction}} B$ $CH_3CH_2CH_2NH_2$

aniline

(ii) $C_6H_5NH_2 \xrightarrow[273\,K]{NaNO_2/HCl} A \xrightarrow[OH^-]{C_6H_5NH_2} B$



Coupling

p-aminoazobenzene

yellow dye

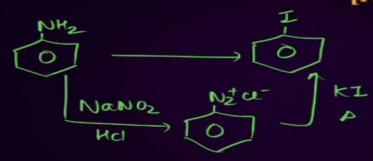
[Delhi 2017, 17(C)]

Q

QUESTION

How can you convert aniline to iodobenzene?

[Foreign 2013; Delhi 2011; DoE]



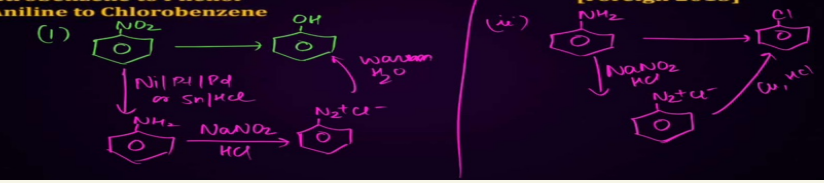
QUESTION

How will you bring about the following conversions?

(i) Nitrobenzene to Phenol

(ii) Aniline to Chlorobenzene

[Foreign 2013]

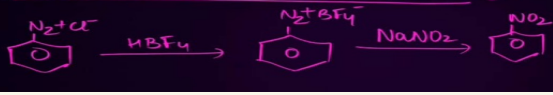


QUESTION

State the reactions and reaction conditions for the following conversions:

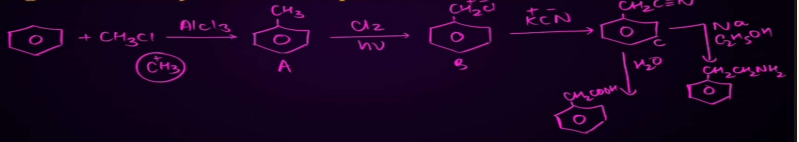
(i) Aniline to benzene diazonium chloride

(ii) Benzene diazonium chloride to nitrobenzene



QUESTION

When benzene reacts with CH_3Cl in presence of $AlCl_3$ to give 'A' (C_7H_8). 'A' reacts with 1 mole of Cl_2 in presence of sun light to form 'B' (C_7H_7Cl). 'B' on reaction with KCN to form 'C'. 'C' on hydrolysis gives 'D'. 'C' on reduction with Na and C_2H_5OH gives 'E'. Identify 'A' to 'e' and explain the reactions:



QUESTION

Give the structures of A, B and C in the following reactions:

(i) $CH_3Br \xrightarrow{KCN} A \xrightarrow{LiAlH_4} B \xrightarrow[273\,K]{HNO_2} C$

CH_3CN $CH_3CH_2NH_2$ $CH_3CH_2N_2^+Cl^-$ $\xrightarrow{H_2O}$ CH_3CH_2OH

(ii) $CH_3COOH \xrightarrow{NH_3} A \xrightarrow{Br_2 + KOH} B \xrightarrow{CHCl_3 + NaOH} C$

CH_3CONH_2 CH_3NH_2 CH_3NC

QUESTION

(a) Write the structures of main products when benzene diazonium chloride ($C_6H_5N_2^+Cl^-$) reacts with the following reagents:

(i) HBF_4/Δ

(ii) Cu/HBr

(b) Write the structures of A, B and C in the following reactions:

(i) $C_6H_5NO_2 \xrightarrow{Sn/HCl} A \xrightarrow[273\,K]{NaNO_2+HCl} B \xrightarrow[\Delta]{H_2O} C$

(ii) $CH_3Cl \xrightarrow{KCN} A \xrightarrow{LiAlH_4} B \xrightarrow[273\,K]{HNO_2} C$

[Foreign 2015]

- describe amines as derivatives of ammonia having a pyramidal structure;
- classify amines as primary, secondary and tertiary;
- name amines by common names and IUPAC system;
- describe some of the important methods of preparation of amines;
- explain the properties of amines;
- distinguish between primary, secondary and tertiary amines;
- describe the method of preparation of diazonium salts and their importance in the synthesis of a series of aromatic compounds including azo dyes.

"The chief commercial use of amines is as intermediates in the synthesis of medicines and fibres".

Amines constitute an important class of organic compounds derived by replacing one or more hydrogen atoms of ammonia molecule by alkyl/aryl group(s). In nature, they occur among proteins, vitamins, alkaloids and hormones. Synthetic examples include polymers, dye stuffs and drugs. Two biologically active compounds, namely adrenaline and ephedrine, both containing secondary amine group, are used to increase blood pressure. Novocain, a synthetic amino compound, is used as an anaesthetic in dentistry. Benadryl, a well known antihistaminic drug also contains tertiary amino group. Quaternary ammonium salts are used as surfactants. Diazonium salts are intermediates in the preparation of a variety of aromatic compounds including dyes. In this Unit, you will learn about amines and diazonium salts.

I. AMINES

Amines can be considered as derivatives of ammonia, obtained by replacement of one, two or all the three hydrogen atoms by alkyl and/or aryl groups.

For example:

$CH_3–NH_2$, $C_6H_5–NH_2$, $CH_3–NH–CH_3$, $CH_3–N(CH_3)_2$

9.1 Structure of Amines

Like ammonia, nitrogen atom of amines is trivalent and carries an unshared pair of electrons. Nitrogen orbitals in amines are therefore, sp^3 hybridised and the geometry of amines is pyramidal. Each of the three sp^3 hybridised orbitals of nitrogen overlap with orbitals of hydrogen or carbon depending upon the composition of the amines. The fourth orbital of nitrogen in all amines contains an unshared pair of electrons. Due to the presence of unshared pair of electrons, the angle C–N–E, (where E is

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- distinguish between primary, secondary and tertiary amines;
- describe the method of preparation of diazonium salts and their importance in the synthesis of a series of aromatic compounds including azo dyes.

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$R\ddot{N}H_2$

$\sigma + \rho = 4$ (sp^3)

$R\text{-}\ddot{N}H\text{-}R$

C or H) is less than 109.5°; for instance, it is 108° in case of trimethylamine as shown in Fig. 9.1.

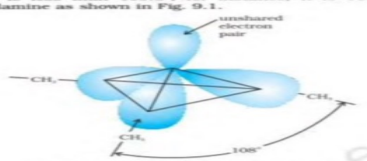


Fig. 9.1 Pyramidal shape of trimethylamine

9.2 Classification

Amines are classified as primary (1°), secondary (2°) and tertiary (3°) depending upon the number of hydrogen atoms replaced by alkyl or aryl groups in ammonia molecule. If one hydrogen atom of ammonia is replaced by R or Ar , we get RNH_2 or $ArNH_2$, a primary amine (1°). If two hydrogen atoms of ammonia or one hydrogen atom of R-NH_2 are replaced by another alkyl/aryl(R') group, what would you get? You get R-NHR', secondary amine. The second alkyl/aryl group may be same or different. Replacement of another hydrogen atom by alkyl/aryl group leads to a tertiary amine. Amines are said to be 'simple' when all the alkyl or aryl groups are the same, and 'mixed' when they are different.

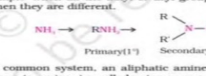
$NH_3 \rightarrow RNH_2 \rightarrow R_2NH \rightarrow R_3N$

Primary(1°) Secondary(2°) Tertiary(3°)

9.3 Nomenclature

In common system, an aliphatic amine is named by prefixing alkyl

aryl groups in ammonia molecule. If one hydrogen atom of ammonia is replaced by R or Ar , we get RNH_2 or $ArNH_2$, a primary amine (1°). If two hydrogen atoms of ammonia or one hydrogen atom of R-NH_2 are replaced by another alkyl/aryl(R') group, what would you get? You get R-NHR', secondary amine. The second alkyl/aryl group may be same or different. Replacement of another hydrogen atom by alkyl/aryl group leads to the formation of tertiary amine. Amines are said to be 'simple' when all the alkyl or aryl groups are the same, and 'mixed' when they are different.



Q.3 Nomenclature

In common system, an aliphatic amine is named by prefixing alkyl group to amine, i.e., alkylamine as one word (e.g., methylamine). In secondary and tertiary amines, when two or more groups are the same, the prefix di or tri is appended before the name of alkyl group. In IUPAC system, primary amines are named as alkanamines. The name is derived by replacement of 'e' of alkane by the word amine. For example, CH_3NH_2 is named as methanamine. In case, more than one amino group is present at different positions in the parent chain, their positions are specified by giving numbers to the carbon atoms bearing $-NH_2$ groups and suitable prefix such as di, tri, etc. is attached to the amine. The letter 'e' of the suffix of the hydrocarbon part is retained. For example, $H_2N-CH_2-CH_2-NH_2$ is named as ethane-1, 2-diamine.

To name secondary and tertiary amines, we use locant N to designate substituent attached to a nitrogen atom. For example, $CH_3NHCH_2CH_3$ is

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Amine	Common name	IUPAC name
$CH_3-CH_2-NH_2$	Ethylamine	Ethanamine
$CH_3-CH_2-CH_2-NH_2$	n-Propylamine	Propan-1-amine
$CH_3-CH(NH_2)-CH_3$	Isopropylamine	Propan-2-amine
$CH_3-NH-CH_2-CH_3$	Ethylmethylamine	N-Methylethanamine
$CH_3-N(CH_3)-CH_3$	Trimethylamine	N,N-Dimethylmethanamine
$C_2H_5-N(C_2H_5)-CH_2-CH_2-CH_3$	N,N-Diethylbutylamine	N,N-Diethylbutan-1-amine
$NH_2-CH_2-CH=CH_2$	Allylamine	Prop-2-en-1-amine
$NH_2-(CH_2)_6-NH_2$	Hexamethylenediamine	Hexane-1,6-diamine
	Aniline	Aniline or Benzenamine
	o-Toluidine	2-Methylaniline
	p-Bromoaniline	4-Bromobenzenamine or 4-Bromoaniline
	N,N-Dimethylaniline	N,N-Dimethylbenzenamine

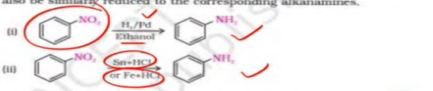
Benzamine ✗
Benzoamine ✗

Q.4 Preparation of Amines

Amines are prepared by the following methods:

1. *Reduction of nitro compounds*

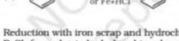
Nitro compounds are reduced to amines by passing hydrogen gas in the presence of finely divided nickel, palladium or platinum and also by reduction with metals in acidic medium. Nitroalkanes can also be similarly reduced to the corresponding alkanamines.



Reduction with iron scrap and hydrochloric acid is preferred because $FeCl_2$ formed gets hydrolysed to release hydrochloric acid during the reaction. Thus, only a small amount of hydrochloric acid is required to initiate the reaction.

2. *Ammonolysis of alkyl halides*

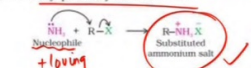
You have read (Unit 6, Class XII) that the carbon - halogen bond in alkyl or benzyl halides can be easily cleaved by a nucleophile. Hence, an alkyl or benzyl halide on reaction with an ethanolic solution of ammonia undergoes nucleophilic substitution reaction in which the halogen atom is replaced by an amino (–NH_2) group. This process of cleavage of the C–X bond by ammonia molecule is known as ammonolysis. The reaction is carried out in a sealed tube at 373 K. The primary amine thus obtained behaves as a nucleophile and can further react with alkyl halide to form secondary and tertiary amines, and finally quaternary ammonium salt.



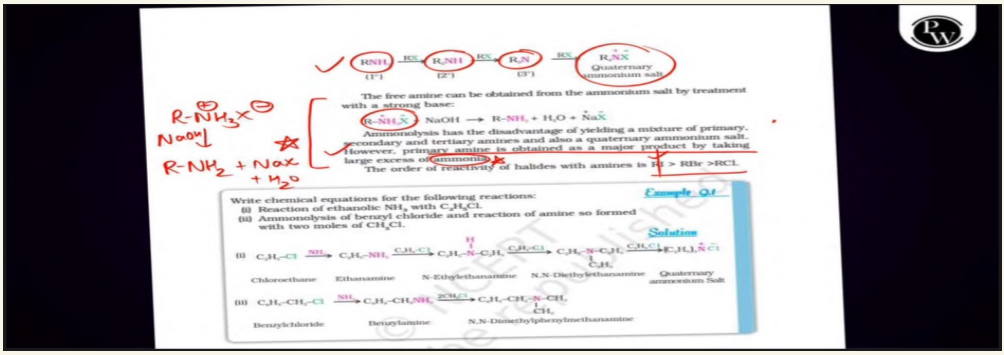
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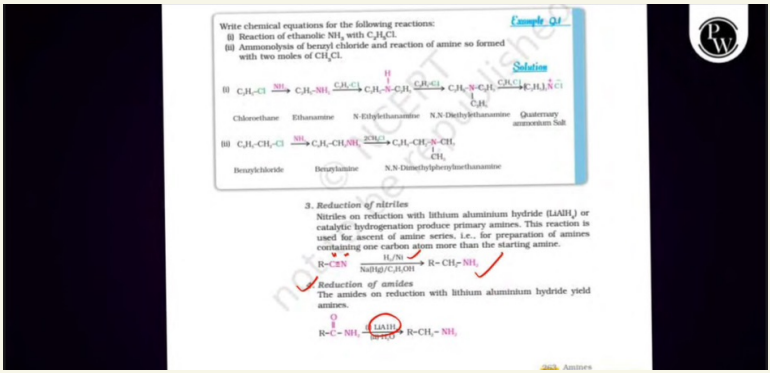
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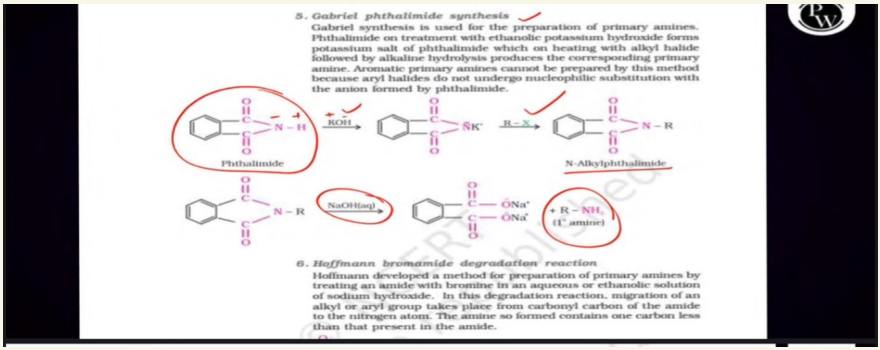
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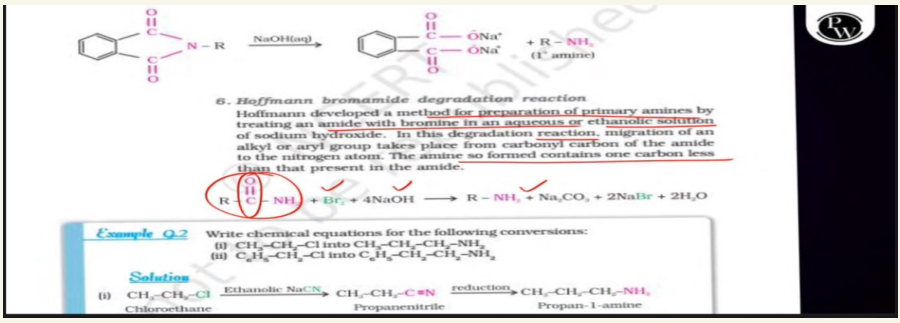


+loving


RNH2
R2NH
R3N
R4N+X-
Quaternary ammonium salt
The free amine can be obtained from the ammonium salt by treatment with a strong base:
Ammonolysis has the disadvantage of yielding a mixture of primary, secondary and tertiary amines and also a quaternary ammonium salt. However, primary amine is obtained as a major product by taking large excess of ammonia.
The order of reactivity of halides with amines is RI > RBr >RCl.
Example Q.1
Write chemical equations for the following reactions:
(i) Reaction of ethanolic NH3 with C2H5Cl.
(ii) Ammonolysis of benzyl chloride and reaction of amine so formed with two moles of CH3Cl.
Solution
Chloroethane
Ethanamine
N-Ethylethanamine
N,N-Diethylethanamine
Quaternary ammonium Salt
Benzylchloride
Benzylamine
N,N-Dimethylphenylmethanamine


Write chemical equations for the following reactions:
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Chloroethane
Ethanamine
N-Ethylethanamine
N,N-Diethylethanamine
Quaternary ammonium Salt
Benzylchloride
Benzylamine
N,N-Dimethylphenylmethanamine
3. Reduction of nitriles
Nitriles on reduction with lithium aluminium hydride (LiAlH4) or catalytic hydrogenation produce primary amines. This reaction is used for ascent of amine series, i.e., for preparation of amines containing one carbon atom more than the starting amine.
Reduction of amides
The amides on reduction with lithium aluminium hydride yield amines.
265 Amines


5. Gabriel phthalimide synthesis
Gabriel synthesis is used for the preparation of primary amines. Phthalimide on treatment with ethanolic potassium hydroxide forms potassium salt of phthalimide which on heating with alkyl halide followed by alkaline hydrolysis produces the corresponding primary amine. Aromatic primary amines cannot be prepared by this method because aryl halides do not undergo nucleophilic substitution with the anion formed by phthalimide.
Phthalimide
N-Alkylphthalimide
6. Hoffmann bromamide degradation reaction
Hoffmann developed a method for preparation of primary amines by treating an amide with bromine in an aqueous or ethanolic solution of sodium hydroxide. In this degradation reaction, migration of an alkyl or aryl group takes place from carbonyl carbon of the amide to the nitrogen atom. The amine so formed contains one carbon less than that present in the amide.


6. Hoffmann bromamide degradation reaction
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Example Q.2
Write chemical equations for the following conversions:
(i) CH3–CH2–Cl into CH3–CH2–CH2–NH2
(ii) C6H5–CH2–Cl into C6H5–CH2–CH2–NH2
Solution
Chloroethane
Propanenitrile
Propan-1-amine

Intext Question

9.3 How will you convert
(i) Benzene into aniline (ii) Benzene into N, N-dimethylaniline
(iii) $Cl-(CH_2)_4-Cl$ into hexan-1,6-diamine?

9.5 Physical Properties

The lower aliphatic amines are gases with fishy odour. Primary amines with three or more carbon atoms are liquid and still higher ones are solid. Aniline and other arylamines are usually colourless but get coloured on storage due to atmospheric oxidation.

Lower aliphatic amines are soluble in water because they can form hydrogen bonds with water molecules. However, solubility decreases with increase in molar mass of amines due to increase in size of the hydrophobic alkyl part. Higher amines are essentially insoluble in water. Considering the electronegativity of nitrogen of amine and oxygen of alcohol as 3.0 and 3.5 respectively, you can predict the pattern of solubility of amines and alcohols in water. Out of butan-1-ol and butan-1-amine, which will be more soluble in water and why? Amines are soluble in organic solvents like alcohol, ether and benzene. You may remember that alcohols are more polar than amines and form stronger intermolecular hydrogen bonds than amines.

Primary and secondary amines are engaged in intermolecular association due to hydrogen bonding between nitrogen of one and hydrogen of another molecule. This intermolecular association is more in primary amines than in secondary amines as there are two hydrogen atoms available for hydrogen bond formation in it. Tertiary amines do not have intermolecular association due to the absence of hydrogen atom available for hydrogen bond formation. Therefore, the order of boiling points of isomeric amines is as follows:

Primary > Secondary > Tertiary

Intermolecular hydrogen bonding in primary amines is shown in Fig. 9.2.

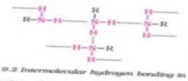


Fig. 9.2 Intermolecular hydrogen bonding in primary amines

Boiling points of amines, alcohols and alkanes of almost the same molar mass are shown in Table 9.2.

Table 9.2: Comparison of Boiling Points of Amines, Alcohols and Alkanes of Similar Molecular Masses

Sl. No.	Compound	Molar mass	b.p./K
1.	$n\text{-}C_4H_9NH_2$	73	350.8
2.	$(C_2H_5)_2NH$	73	329.3
3.	$C_2H_5NCH_3)_2$	73	310.5
4.	$C_2H_5CHCH_3)_2$	72	300.9
5.	$n\text{-}C_4H_9OH$	74	390.3

9.6 Chemical Reactions

Difference in electronegativity between nitrogen and hydrogen atoms and the presence of unshared pair of electrons over the nitrogen atom makes amines reactive. The number of hydrogen atoms attached to nitrogen atom also decides the course of reaction of amines; that is why primary ($-NH_2$), secondary ($>N-H$) and tertiary amines ($>N-$) differ in many reactions. Moreover, amines behave as nucleophiles due to the presence of unshared electron pair. Some of the reactions of amines are described below:

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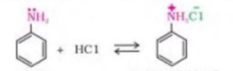
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1. Basic character of amines

Amines, being basic in nature, react with acids to form salts.

$$R-\ddot{N}H_2 + HX \rightleftharpoons R-\overset{+}{N}H_3\,\bar{X} \quad \text{(Salt)}$$



Aniline Anilinium chloride

Amine salts on treatment with a base like NaOH, regenerate the parent amine.

$$R\overset{+}{N}H_3\,\bar{X} + \bar{O}H \longrightarrow R\ddot{N}H_2 + H_2O + \bar{X}$$

Amine salts are soluble in water but insoluble in organic solvents like ether. This reaction is the basis for the separation of amines from the non basic organic compounds insoluble in water.

The reaction of amines with mineral acids to form ammonium salts shows that these are basic in nature. Amines have an unshared pair of electrons on nitrogen atom due to which they behave as Lewis base. Basic character of amines can be better understood in terms of their K_b and pK_b values as explained below:

$$R-\ddot{N}H_2 + H_2O \rightleftharpoons R-\overset{+}{N}H_3 + \bar{O}H$$

$$K = \frac{[R-\overset{+}{N}H_3][\bar{O}H]}{[R-\ddot{N}H_2][H_2O]}$$

$$\text{or } K[H_2O] = \frac{[R-\overset{+}{N}H_3][\bar{O}H]}{[R-\ddot{N}H_2]}$$

$$\text{or } K_b = \frac{[R-\overset{+}{N}H_3][\bar{O}H]}{[R-\ddot{N}H_2]}$$

$$pK_b = -\log K_b$$

Larger the value of K_b or smaller the value of pK_b, stronger is the base. The pK_b values of few amines are given in Table 9.3.

pK_b value of ammonia is 4.75. Aliphatic amines are stronger bases than ammonia due to +I effect of alkyl groups leading to high electron density on the nitrogen atom. Their pK_b values lie in the range of 3 to 4.22. On the other hand, aromatic amines are weaker bases than ammonia due to the electron withdrawing nature of the aryl group.

$R\overset{+}{N}H_3\bar{X}$ + NaOH → $NaX + H_2O + RNH_2$

$pH = -\log[H^+]$

$H^+ \uparrow \; pH \downarrow$

$pK_b = -\log K_b$

$K_b \uparrow \; pK_b \downarrow$

$K_b = \frac{[R\overset{+}{N}H_3][\bar{O}H]}{[RNH_2][H_2O]}$

donate ↑ of e^-

$K_b \uparrow$ (basic strength) $\; pK_b \downarrow$

$-I < +I < +M$

Aliphatic amine

NH_3

Aromatic amine

CH_3-NH_2 > NH_3 > $C_6H_5NH_2$

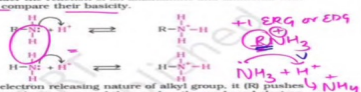
+I 1 2 3 -I

Kb 1 > 2 > 3

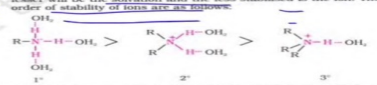
pKb 1 < 2 < 3

(a) *Alkanamines versus ammonia*

Let us consider the reaction of an alkanamine and ammonia with a proton to compare their basicity.



Due to the electron releasing nature of alkyl group, it (R) pushes electrons towards nitrogen and thus makes the unshared electron pair more available for sharing with the proton of the acid. Moreover, the substituted ammonium ion formed from the amine gets stabilised due to dispersal of the positive charge by the +I effect of the alkyl group. Hence, alkylamines are stronger bases than ammonia. Thus, the basic nature of aliphatic amines should increase with increase in the number of alkyl groups. This trend is followed in the gaseous phase. The order of basicity of amines in the gaseous phase follows the expected order: tertiary amine > secondary amine > primary amine > NH_3. The trend is not regular in the aqueous state as evident by their pK_b values given in Table 9.3. In the aqueous phase, the substituted ammonium cations get stabilised not only by electron releasing effect of the alkyl group (+I) but also by solvation with water molecules. The greater the size of the ion, lesser will be the solvation and the less stabilised is the ion. The order of stability of ions are as follows:



1° 2° 3°

Decreasing order of extent of H-bonding in water and order of stability of ions by solvation.

Due to the electron releasing nature of alkyl group, it (R) pushes electrons towards nitrogen and thus makes the unshared electron pair more available for sharing with the proton of the acid. Moreover, the substituted ammonium ion formed from the amine gets stabilised due to dispersal of the positive charge by the +I effect of the alkyl group. Hence, alkylamines are stronger bases than ammonia. Thus, the basic nature of aliphatic amines should increase with increase in the number of alkyl groups. This trend is followed in the gaseous phase. The order of basicity of amines in the gaseous phase follows the expected order: tertiary amine > secondary amine > primary amine > NH_3. The trend is not regular in the aqueous state as evident by their pK_b values given in Table 9.3. In the aqueous phase, the substituted ammonium cations get stabilised not only by electron releasing effect of the alkyl group (+I) but also by solvation with water molecules. The greater the size of the ion, lesser will be the solvation and the less stabilised is the ion. The order of stability of ions are as follows:

Decreasing order of extent of H-bonding in water and order of stability of ions by solvation.

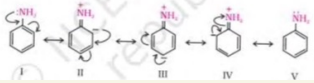
Greater is the stability of the substituted ammonium cation, stronger should be the corresponding amine as a base. Thus, the order of basicity of aliphatic amines should be: primary > secondary > tertiary, which is opposite to the inductive effect based order. Secondly, when the alkyl group is small, like $-CH_3$ group, there is no steric hindrance to H-bonding. In case the alkyl group is bigger than CH_3 group, there will be steric hinderance to H-bonding. Therefore, the change of nature of the alkyl group, e.g., from $-CH_3$ to $-C_2H_5$ results in change of the order of basic strength. Thus, there is a subtle interplay of the inductive effect, solvation effect and steric hinderance of the alkyl group which decides the basic strength of alkyl amines in the aqueous state. The order of basic strength in case of methyl substituted amines and ethyl substituted amines in aqueous solution is as follows:

ethyl: $(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5NH_2 > NH_3$

methyl: $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$

(b) *Arylamines versus ammonia*

pK_b value of aniline is quite high. Why is it so? It is because in aniline or other arylamines, the $-NH_2$ group is attached directly to the benzene ring. It results in the unshared electron pair on nitrogen atom to be in conjugation with the benzene ring and thus making it less available for protonation. If you write different resonating structures of aniline, you will find that aniline is a resonance hybrid of the following five structures.



I II III IV V

donate ℓp of e^-
accept H^+

Aliphatic > NH_3 > aromatic

$3^\circ > 2^\circ > 1^\circ$ ✓

$2^\circ > 1^\circ > 3^\circ$

$(CH_3)NH_2$

+I EDG

or ERG

substituted amines in aqueous solution is as follows:

$(C_2H_5)_2NH > (C_2H_5)_3N > C_2H_5NH_2 > NH_3$

$(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$

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I II III IV V

On the other hand, anilinium ion obtained by accepting a proton can have only two resonating structures (kekule).

I II

We know that greater the number of resonating structures, greater is the stability. Thus you can infer that aniline (five resonating structures) is more stable than anilinium ion. Hence, the proton acceptability or the basic nature of aniline or other aromatic amines would be less than that of ammonia. In case of substituted aniline, it is observed that electron releasing groups like $-OCH_3$, $-CH_3$ increase

a base stronger than the amine, like pyridine, which removes HCl so formed and shifts the equilibrium to the right hand side.

Ethanamine — N-Ethylethanamide

N-Ethylethanamine — N,N-Diethylethanamide

Benzenamine — Ethanoic anhydride — N-Phenylethanamide or Acetanilide

Amines also react with benzoyl chloride (C_6H_5COCl). This reaction is known as benzoylation.

$CH_3NH_2 + C_6H_5COCl \rightarrow CH_3NHCOC_6H_5 + HCl$

Methanamine — Benzoyl chloride — N – Methylbenzamide

What do you think is the product of the reaction of amines with carboxylic acids ? They form salts with amines at room temperature.

4. Carbylamine reaction

Aliphatic and aromatic primary amines on heating with chloroform and ethanolic potassium hydroxide form isocyanides or carbylamines which are foul smelling substances. Secondary and tertiary amines do not show this reaction. This reaction is known as **carbylamine reaction** or **isocyanide test** and is used as a test for primary amines.

$R-NH_2 + CHCl_3 + 3KOH \xrightarrow{Heat} R-NC + 3KCl + 3H_2O$

5. Reaction with nitrous acid

Three classes of amines react differently with nitrous acid which is prepared *in situ* from a mineral acid and sodium nitrite.

(a) Primary aliphatic amines react with nitrous acid to form aliphatic diazonium salts which being unstable, liberate nitrogen gas quantitatively and alcohols. Quantitative evolution of nitrogen is used in estimation of amino acids and proteins.

$R-NH_2 + HNO_2 \xrightarrow{NaNO_2 + HCl} [R-\overset{+}{N}_2Cl] \xrightarrow{H_2O} ROH + N_2 + HCl$

(b) Aromatic amines react with nitrous acid at low temperatures (273-278 K) to form diazonium salts, a very important class of compounds used for synthesis of a variety of aromatic compounds discussed in Section 9.7.

$C_6H_5-NH_2 \xrightarrow[273-278\ K]{NaNO_2 + 2HCl} C_6H_5-\overset{+}{N}_2Cl^- + NaCl + 2H_2O$

Aniline — Benzenediazonium chloride

Secondary and tertiary amines react with nitrous acid in a different manner.

6. Reaction with arylsulphonyl chloride

Benzenesulphonyl chloride ($C_6H_5SO_2Cl$), which is also known as **Hinsberg's reagent**, reacts with primary and secondary amines to form sulphonamides.

(a) The reaction of benzenesulphonyl chloride with primary amine yields N-ethylbenzenesulphonyl amide.

N-Ethylbenzenesulphonamide (soluble in alkali)

The hydrogen attached to nitrogen in sulphonamide is strongly acidic due to the presence of strong electron withdrawing sulphonyl group. Hence, it is soluble in alkali.

(b) In the reaction with secondary amine, N,N-diethylbenzenesulphonamide is formed.

N,N-Diethylbenzenesulphonamide

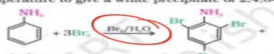
Since N, N-diethylbenzene sulphonamide does not contain any hydrogen atom attached to nitrogen atom, it is not acidic and hence insoluble in alkali.

(c) Tertiary amines do not react with benzenesulphonyl chloride. This property of amines reacting with benzenesulphonyl chloride in a different manner is used for the distinction of primary, secondary and tertiary amines and also for the separation of a mixture of amines. However, these days benzenesulphonyl chloride is replaced by *p*-toluenesulphonyl chloride.

7. ***Electrophilic substitution***

You have read earlier that aniline is a resonance hybrid of five structures. Where do you find the maximum electron density in these structures? *Ortho-* and *para*-positions to the $-NH_2$ group become centres of high electron density. Thus $-NH_2$ group is *ortho* and *para* directing and a powerful activating group.

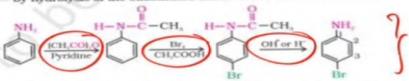
(a) Bromination: Aniline reacts with bromine water at room temperature to give a white precipitate of 2,4,6-tribromoaniline.



Aniline 2,4,6-Tribromoaniline

The main problem encountered during electrophilic substitution reactions of aromatic amines is that of their very high reactivity. Substitution tends to occur at *ortho-* and *para*-positions. If we have to prepare monosubstituted aniline derivative, how can the activating effect of $-NH_2$ group be controlled ? This can be

The main problem encountered during electrophilic substitution reactions of aromatic amines is that of their very high reactivity. Substitution tends to occur at *ortho-* and *para*-positions. If we have to prepare monosubstituted aniline derivative, how can the activating effect of $-NH_2$ group be controlled ? This can be done by protecting the $-NH_2$ group by acetylation with acetic anhydride, then carrying out the desired substitution followed by hydrolysis of the substituted amide to the substituted amine.



Aniline *N*-Phenylethanamide (Acetanilide) (Major) 4-Bromoaniline

The lone pair of electrons on nitrogen of acetanilide interacts with oxygen atom due to resonance as shown below:

